From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE **JEE (Main), 2014**

CONCEPT RECAPITULATION TEST - II

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

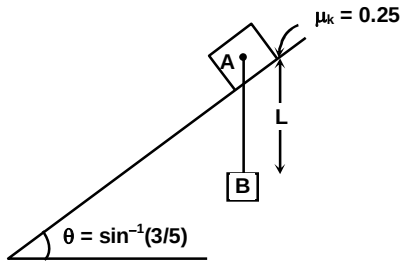
PART – I

SECTION – A

Single Correct Choice Type

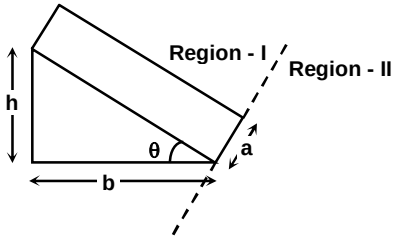
This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Two similar blocks A and B are riveted rigidly with a rigid rod of length L such that block A can slide along an inclined plane as shown in the figure. If masses of the both blocks and the rod are each equal to m and kinetic friction between A and inclined plane is $\mu_k = 0.25$ then the magnitude of acceleration of centre of mass of the block rod system is



(A) 6 m/s^2 (B) 4 m/s^2
(C) 5 m/s^2 (D) 2 m/s^2

2. A wedge of mass M and dimensions as shown in the figure is lying completely inside a region (I) of zero pressure. The pressure inside region II is 1 Pa. The surfaces are smooth. The wedge is imparted an initial velocity v_0 so that it just travels a distance b inside region II before finally coming to rest. The value of v_0 is

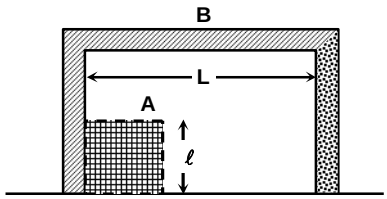


(A) $\sqrt{\frac{abh}{M}}$ (B) $\frac{3}{4}\sqrt{\frac{abh}{M}}$

(C) $\frac{4}{3}\sqrt{\frac{abh}{M}}$ (D) $\sqrt{\frac{4}{3}\frac{abh}{M}}$

Space for Rough work

3. A block A of mass m and a box B of same mass are lying on a horizontal surface as shown in the above figure. The coefficient of kinetic friction between the block and horizontal surface is μ_A, while that between the box and surface is μ_B. Given that $\mu_B > \mu_A$. At t = 0, the position is as shown in the figure and both are simultaneously given a velocity v_0 to the right. The time after which block will collide with the right end of the box is

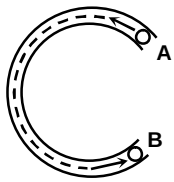


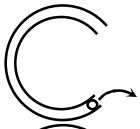
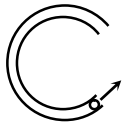
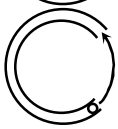
(A) $\sqrt{\frac{2(L-\ell)}{(\mu_B-\mu_A)g}}$ (B) $\sqrt{\frac{2(L-\ell)}{(\mu_B+\mu_A)g}}$

(C) $\sqrt{\frac{4(L+\ell)}{(\mu_B-\mu_A)g}}$ (D) $\sqrt{\frac{4(L-\ell)}{(\mu_B-\mu_A)g}}$

4. It is observed that only 0.39% of the original radioactive sample remains undecayed after eight hours. Hence choose the incorrect statement.
(A) The half life of that substance is 1 hr.
(B) The mean life of that substance is 1/ln 2 hr
(C) The decay constant of the substance is ln 2 hr^{-1}
(D) The half life of the substance is 1/2 hr

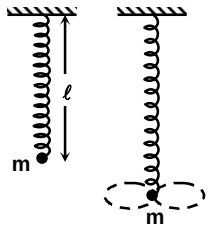
5. The figure shows a hollow circular tube fixed to a frictionless horizontal table. It is the top view of the arrangement. A ball is shot into end A of the tube to leave the other end B at high speed. The balls diameter is slightly less than the internal diameter of the tube. Which of the paths below will the ball follow on the table after it leaves the tube?



(A) 
(B) 
(C) 
(D) None of these

Space for Rough work

6. A mass m on the end of an ideal spring stretches the spring to a length ℓ when at rest. The mass is now set into motion so that it executes up and down vibrations while swinging back and forth as a simple pendulum. The mass moves in a figure eight pattern in a vertical plane a shown in the figure. The force constant of the spring assuming small oscillations is



(A) $\frac{mg}{\ell}$ (B) $\frac{2mg}{k}$

(C) $\frac{4mg}{\ell}$ (D) $\frac{mg}{2\ell}$

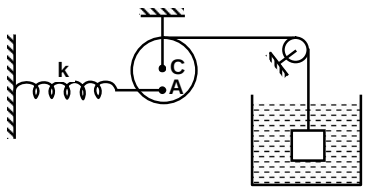
7. When an ultrasonic waves travels from air into water,
(A) waves bends towards the normal
(B) frequency of waves changes from air to water.
(C) speed of ultrasonic wave is greater in water than that in air.
(D) speed of ultrasonic wave is less in water than that in air.

8. A narrow tunnel is made through the earth of mass M and radius R from the centre to the surface. The speed (in m/s) with which a particle should be projected from the centre of the earth through the tunnel so that it escapes to the space is

(A) $\sqrt{2gR}$ (B) $\sqrt{gR}$

(C) $\sqrt{3gR}$ (D) $2\sqrt{gR}$

9. A cubical block of volume v and density 3ρ is placed inside a liquid of density ρ and attached to a spring of spring constant k as shown in the figure. Assuming ideal spring and pulley and spring is attached at A which is at R/2 from centre. The compression in the spring at equilibrium is

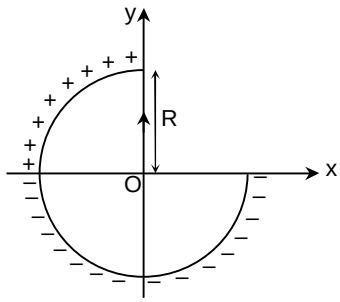


(A) $\frac{4\rho vg}{k}$ (B) $\frac{2\rho vg}{k}$

(C) $\frac{3\rho vg}{k}$ (D) $\frac{\rho vg}{k}$

Space for Rough work

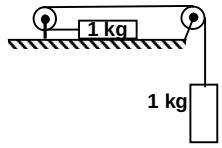
10. A conductor having uniform linear charge density λ is placed in circular form in 3 quadrants with different polarities as shown in the figure. Let $\vec{E}_0$ be the electric field at O. Then the angle made by $\vec{E}_0$ with the line along y-axis is



(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) $\tan^{-1}(1)$

(C) $\tan^{-1}\left(\frac{1}{3}\right)$ (D) None

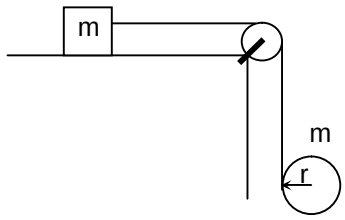
11. The pulley and the string are light and all the surfaces are frictionless. The system is initially held at rest and The string is just taut. The 1kg block on the plane is suddenly imparted a velocity of 5m/s to the left, and the system is simultaneously released. Find the displacement of the 1kg block (on the plane) after 3s. [Take g = 10 m/s^2]



(A) 10 m (B) 20 m

(C) 30 m (D) 40 m

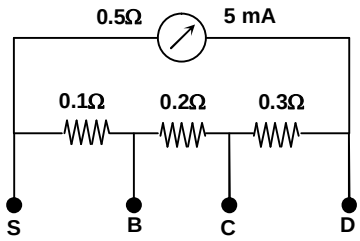
12. A light thread is wound on a disk of mass *m* and other end of thread is connected to a block of mass *m*, which is placed on a rough ground as shown in diagram. Find the minimum value of coefficient of friction for which block remain at rest:



(A) $\frac{1}{3}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{1}{5}$

13. A milli ammeter of range 5 mA and resistance 0.5 Ω is joined in a circuit as shown. Find the value of current for which meter gives full scale deflection when A & B are used as Terminal:

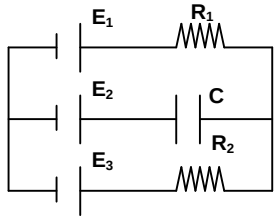


(A) 50 mA (B) 5 mA

(C) 100 mA (D) 1 A

Space for Rough work

14. In the circuit shown, the batteries E_1 = E_2 = 1 V, E_3 = 2.5 V and the resistances R_1 = 10 Ω, R_2 = 20 Ω. The potential difference across the capacitor is



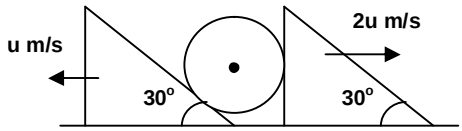
(A) 0.2 V (B) 0.4 V
(C) 0.5 V (D) 1.2 V

15. A horizontal spring mass system of mass M performs simple harmonic motion of amplitude a_0 and time period T_0. When the mass M passes through mean position another stationary mass M sticks to it and both move together. If *a* and T be new amplitude and time period. Then

(A) $a = \frac{a_0}{\sqrt{2}}$ (B) $a = a_0\sqrt{2}$

(C) $T = \frac{T_0}{\sqrt{2}}$ (D) None

16. System is shown in figure. Assume the cylinder remains in contact with the two wedges. The velocity of cylinder is



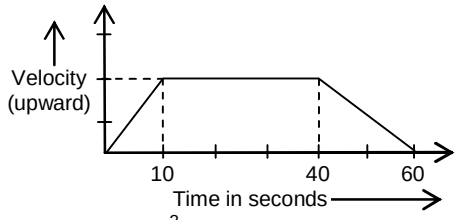
(A) $\sqrt{19-4\sqrt{3}}\,\frac{u}{2}$ m/s (B) $\frac{\sqrt{13}}{2}$ v m/s

(C) $\sqrt{3}$v m/s (D) $\sqrt{7}$u m/s

17. Consider the following statements for a particle moving in an elliptic orbit under the influence of a central force:
(1) The radius vector covers equal area in equal time.
(2) The motion takes place in a plane.
(3) The angular momentum is constant of motion.
Which of options given above are correct?
(A) 1 & 2 only (B) 2 & 3 only
(C) 1 & 3 only (D) all

Space for Rough work

18. A person is standing on a weighing machine placed on the floor of an elevator. The motion of the elevator is shown in the adjacent diagram. The maximum and the minimum weights recorded are 66 kg and 57 kg. The magnitude of the upward accelerations is [Take g = 10 m/s^2.]



(A) 1 m/s^2 (B) 0.5 m/s^2
(C) 2 m/s^2 (D) 0.25 m/s^2

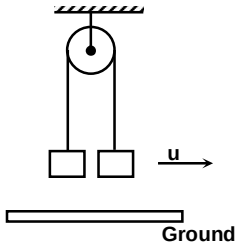
19. When a charge particle of charge $2\mu c$ and mass 1 gm is released at origin in gravity free space having $\vec{E} = E_0\hat{j}$ and $\vec{B} = -B_0\hat{j}$ ($E_0 = 10^3$ N/c and B0 = 5T). Then its speed

(A) depends only of X co-ordinate (B) depends only on Y co-ordinate
(C) depends only on z co-ordinate (D) depends only on X, Y, Z co-ordinate

20. A neutral metallic finite block is placed at large but finite distance from a large charged sheet in the middle space in front of sheet. Then the block will be

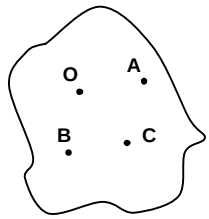
(A) Attracted towards the sheet (B) Repelled away from the sheet
(C) Depend on nature of charge on the sheet (D) Zero force on the block

21. Two equal masses hang on either side of a pulley at the same height from the ground. The mass on the right is given a horizontal speed, after some time.



(A) The mass on the left will be nearer to ground.
(B) The mass on the right will be nearer to ground.
(C) Both the masses will be at equal distance from the ground.
(D) Nothing can be said regarding their positions.

22. O is the centre of mass of a body of mass M as shown in the figure. A, B, C are three different point on the body. OB = 8 cm, OC = 10 cm, BC = 6 cm and OA = 10 cm. Which of the following can be written by using parallel axis theorem? I_0 is the moment of inertia about the axis passing through point O and perpendicular to plane of object.



(A) $I_B = I_C + M(BC)^2$ (B) $I_C = I_B + M(BC)^2$
(C) $I_A = I_0 + M(OB)^2$ (D) None of these

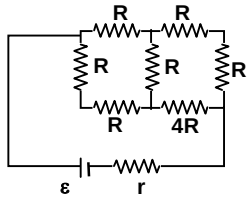
Space for Rough work

23. A solid sphere of radius R, and dielectric constant 'k' has spherical cavity of radius R/4. A point charge q_1 is placed in the cavity. Another charge q_2 is placed outside the sphere at a distance of r from q. Then Coulombic force of interaction between them is found to be 'F_1'. When the same charges are separated by same distance in vacuum then the force of interaction between them is found to be F_2 then

(A) $F_1 = F_2/k$ (B) $F_2 = F_1/k$

(C) $F_1 . F_2 = \frac{1}{k}$ (D) $F_1 = F_2$

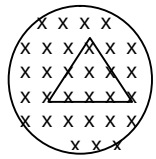
24. The relation between R and r (internal resistance of the battery) for which the power consumed in the external part of the circuit is maximum.



(A) R = r (B) $R = \frac{r}{2}$

(C) R = 2r (D) R = 1.5 r

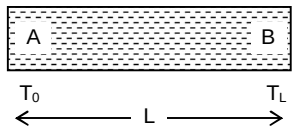
25. An equilateral triangular loop having a resistance R and length of each side 'ℓ' is placed in a magnetic field which is varying at $\frac{dB}{dt}$ = 1 T/s. The induced current in the loop will be



(A) $\frac{\sqrt{3}}{4}\frac{\ell^2}{R}$ (B) $\frac{4}{\sqrt{3}}\frac{\ell^2}{R}$

(C) $\frac{\sqrt{3}}{4}\frac{R}{\ell^2}$ (D) $\frac{4}{\sqrt{3}}\frac{R}{\ell^2}$

26. The temperature of a mono-atomic gas in an uniform container of length 'L' varies linearly from T_0 to T_L as shown in the figure. If the molecular weight of the gas is M_0, then the time taken by a wave pulse in travelling from end A to end B is

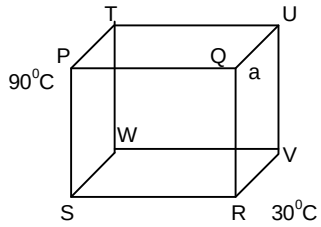


(A) $\frac{2L}{(\sqrt{T_L}+\sqrt{T_0})}\sqrt{\frac{3M}{5R}}$ (B) $\sqrt{\frac{3(T_L - T_0)}{5RM_0L}}$

(C) $\frac{2L}{(\sqrt{T_L}-\sqrt{T_0})}\sqrt{\frac{3M}{5R}}$ (D) $L\sqrt{\frac{M_0}{2R(T_L - T_0)}}$

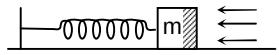
Space for Rough work

27. 12 identical rods made of same material are arranged in the form of a cube. The temperature of 'P' and 'R' are maintained at 90^0 C and 30^0C respectively. Then the temperature of point 'V', when steady state is reached,



(A) 65°C (B) 60°C
(C) 20°C (D) 50°C

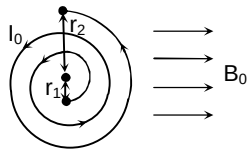
28. A spring block system with mass of block m and spring constant K (all the surfaces of block are perfectly reflecting and smooth) is placed on a smooth horizontal plane as shown in the diagram. A light beam of intensity I is switched on from rightwards. Find the amplitude of oscillations of the block.



(A) $\frac{I}{KC}$ (B) $\frac{2I}{KC}$
(C) $\frac{4I}{KC}$ (D) Zero

29. A rod of mass M and length L is placed on a smooth horizontal table and is hit by a ball moving horizontally and perpendicular to length of rod and sticks to it. Then conservation of angular momentum can be applied
(A) About any point on the rod (B) About a point at the centre of the rod
(C) About end point of the rod (D) None

30. Magnetic force on a spiral carrying current I_0 and placed in magnetic field B_0 parallel to the plane of spiral as shown in diagram, will be nearly (initial point, final point and centre lie on a line)



(A) Zero (B) $I_0B_0(r_2 - r_1)$
(C) $I_0B_0(r_2 + r_1)$ (D) $\frac{I_0B_0r_2}{r_1}$

Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The pressure (P) and density (ρ) of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes from P_1 ρ_1 to $P_2\rho_2$. If $\frac{\rho_1}{\rho_2} = 32$, the value of $\frac{P_1}{P_2}$ is:

(A) 32 (B) $\frac{1}{32}$

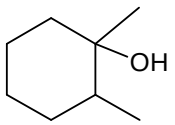
(C) 128 (D) $\frac{1}{164}$

2. 0.2 mol of NH_4Cl are introduced into an empty container of 10 litre and heated to 327°C to attain equilibrium as $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$, K_p = 0.36 atm^2. The quantity of solid NH_4Cl left is:

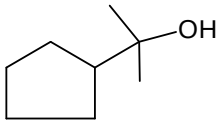
(A) 0.02 mole (B) 0.078 mole

(C) 0.095 mole (D) 0.035 mole

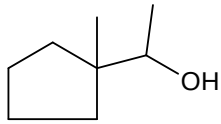
3. Which of the following alcohols will give same alkene on reaction with conc. H_2SO_4?



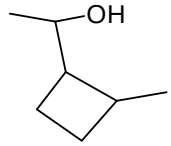
(I)



(II)



(III)



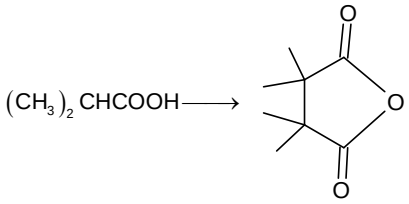
(IV)

(A) I and II (B) II and III

(C) I, II and III (D) II, III and IV

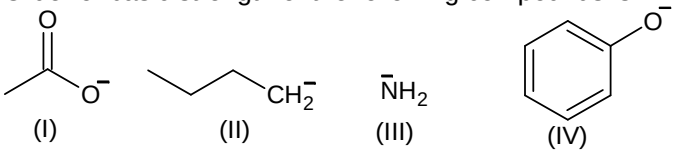
Space for Rough work

4. The correct set of reagents for the following conversion is:



(A) P_4/I_2, Na, conc. H_2SO_4
(B) P_2O_5, $LiAlH_4$
(C) P_4/I_2, Na, P_2O_5/Δ
(D) P_2O_5/Δ, H_2O, P_4/I_2, Na

5. Order of basic strength of the following compounds is:



(I) (II) (III) (IV)

(A) IV > II > I > III
(B) III > II > IV > I
(C) II > III > IV > I
(D) II > III > I > IV

6. When K_2CrO_4 is added to $CuSO_4$ solution there is formation of $CuCrO_4$ and $CuCr_2O_7$. Formation of $CuCr_2O_7$ is due to:
(A) basic nature of $CuSO_4$ solution which converts CrO_4^{2-} to $Cr_2O_7^{2-}$
(B) acidic nature of $CuSO_4$ solution which converts CrO_4^{2-} to $Cr_2O_7^{2-}$
(C) $CuSO_4$ has a typical property of converting CrO_4^{2-} to $Cr_2O_7^{2-}$
(D) none of the above is correct

7. Amorphous boron is extracted from borax by following steps:

$$\text{Borax} \xrightarrow{(X)} H_3BO_3 \xrightarrow{(Y)} B_2O_3 \xrightarrow{(Z)} \text{Boron}$$

What are (X) and (Z)?
(A) H_2SO_4, Al
(B) HCl, C
(C) HCl, Fe
(D) H_2SO_4, Na

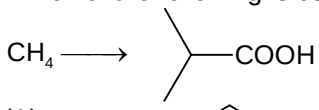
8. Heat of neutralisation of strong acid and strong base under 1 atm and 25°C is – 13.7 kcal. If standard Gibbs energy change for dissociation of water to H^+ and OH^- is – 19.14 kcal, the change in standard entropy for dissociation of water in cal k^{-1} mol^{-1} is:
(A) 18.25
(B) 110.2
(C) - 18.25
(D) none of these

Space for Rough work

9. A metal crystallises in bcc lattice. The % of fraction of edge length not covered by atom is:
(A) 10.4% (B) 13.4%
(C) 12.4% (D) 11.4%

10. $E^0_{Al^{3+}/Al} = -1.66$ V and K_{sp} of $Al(OH)_3$ = 1.0×10^{-33}. The reaction potential of above couple at pH = 14 is:
(A) - 2. 31 V (B) + 2.31 V
(C) + 1.01 V (D) - 1.01 V

11. Which of the following is correct option of reagent for the given conversion?



(A) $Br_2/h\nu$,  $MgCl$, $Br_2/h\nu$, $HCOOH$

(B) $Cl_2/h\nu$,  $MgCl$, $Br_2/h\nu$, KCN/H_3O^+

(C) $Br_2/h\nu$,  $MgCl$, $Br_2/h\nu$, $NaNH_2$, HCN, H_3O^+

(D) $Cl_2/h\nu, CH_3MgBr, Br_2/h\nu, CH_3COOH$

12. Which is correct statements about P_4O_6 and P_4O_{10}?
(A) Both form oxyacids H_3PO_3 and H_3PO_4 respectively.
(B) In P_4O_4 each P is joined to three O and in P_4O_{10} each P is joined to four O atoms.
(C) Both of these
(D) None of these

13. 50 mL of a solution of Na_2CO_3 neutralizes 49.35 mL of 4 N HCl. The reaction is represented as
$CO_3^{2-} + 2H^+ \longrightarrow CO_2 + H_2O$
The density of this Na_2CO_3 solution is 1.25 g mL^{-1}. The percentage of Na_2CO_3 in it is:
(A) 47.7 (B) 37.7
(C) 26.7 (D) 16.7

Space for Rough work

14. In the extraction of copper, the reaction which does not take place in Bessemer converter is:

(A) $2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$ (B) $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

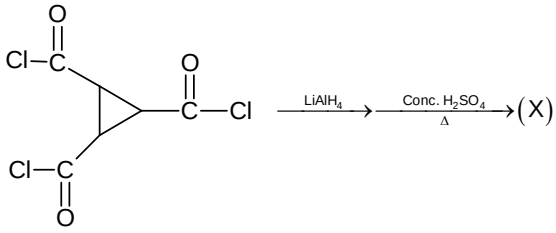
(C) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$ (D) $2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$

15. In the following compounds of manganese what is the distribution of electrons on d-orbitals of manganese?

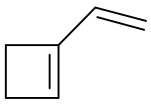
(i) $[Mn(H_2O)_6]^{2+}$ (ii) $[Mn(CN)_6]^{4-}$

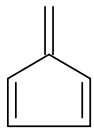
(A) $t_{2g}^3 e_g^2$ in both (B) $t_{2g}^5 e_g^0$ in both

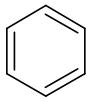
(C) $t_{2g}^3 e^2$ in (i) and $t_{2g}^5 e_g^0$ in (ii) (D) $t_{2g}^5 e_g^0$ in (i) and $t_{2g}^3 e_g^2$ (ii)

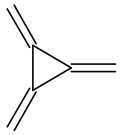
16. 

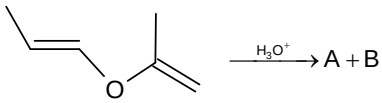
The final product (X) is:

(A) 

(B) 

(C) 

(D) 

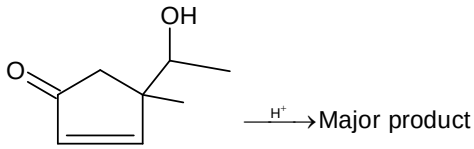
17. 

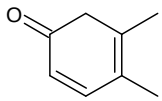
Compounds A and B can be differentiated by

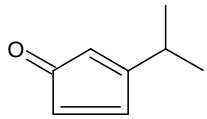
(A) 2, 4-DNP (B) Fehling solution

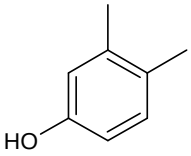
(C) Lucas reagent (D) $NaHSO_3$

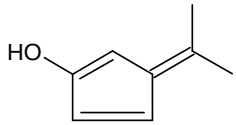
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18. 

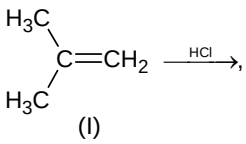
(A) 

(B) 

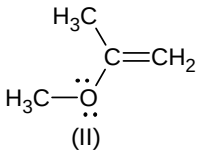
(C) 

(D) 

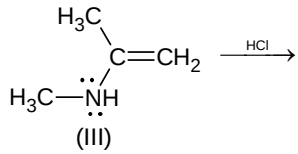
19. Arrange the following reactions ion decreasing order of electrophilic addition reaction:



(I)



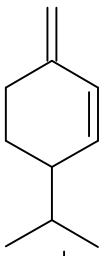
(II)

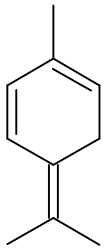


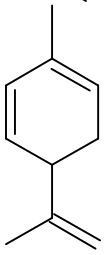
(III)

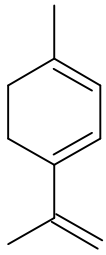
(A) I > II > III
(B) III > II > I
(C) II > III > I
(D) I = II = III

20. Reductive ozonolysis of alkene $C_{10}H_{14}$ gave $CH_3 - CO - CH_3$, $CH_3 - COCHO$ and $OCH - CH_2CO - CHO$ in 1 : 1 : 1 ratio. Which of the following is the most likely structure of alkene.

(A) 

(B) 

(C) 

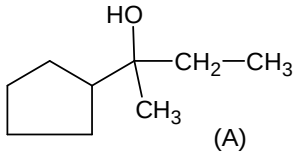
(D) 

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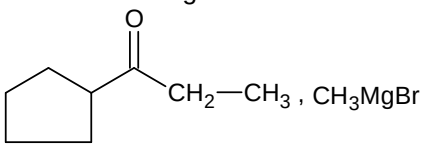
21. Temperature of 4 moles of an ideal gas is raised from 300 K to 350 K. What is value of $\Delta H - \Delta E$ for this process? (R = 8.314 J J $mol^{-1}K^{-1}$)
(A) 0 (B) 415 J
(C) 41.5 (D) 1660 J

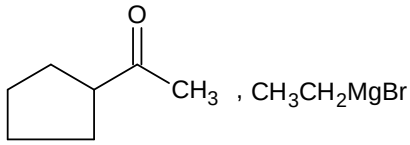
22. K_{sp} of CdS is 8.0×10^{-27} and that of H_2S is 1×10^{-22}. 1×10^{-14} M $CdCl_2$ solution is precipitated on passing H_2S when pH is about
(A) 4 (B) 6
(C) 5 (D) 7

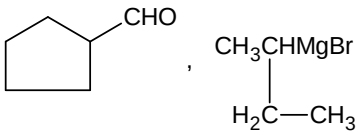
23. For the following equilibrium reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, NO_2 is 50% of total volume at given temperature. Hence, vapour density of equilibrium mixture is:
(A) 34.5 (B) 25.0
(C) 23.0 (D) 20.0

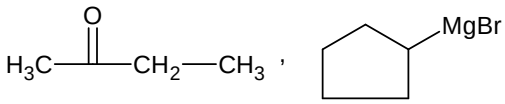
24. To synthesize 

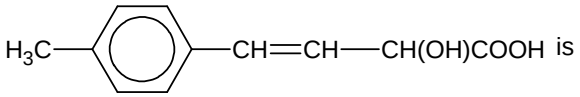
(A)

Which of the following reactants cannot be suitable:
(A) 

(B) 

(C) 

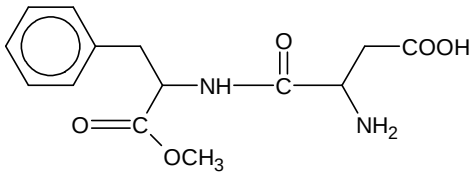
(D) 

25. The number of stereoisomers possible for the compound


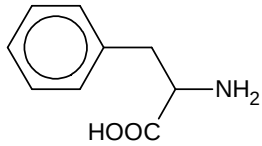
is
(A) 2 (B) 4
(C) 6 (D) 8

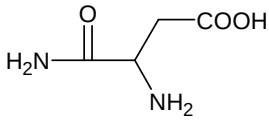
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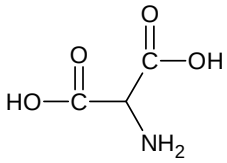
26. Aspartame, an artificial sweetener is peptide and has the following structure:



Free amino acids obtained by the hydrolysis of aspartame are

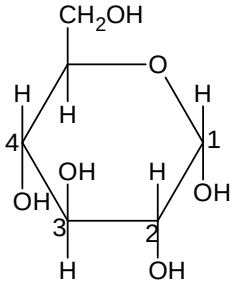
(A) 

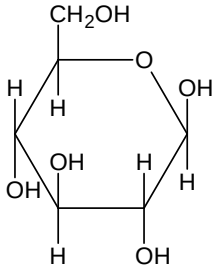
(B) 

(C) 

(D) Both (A) and (C)

27. Haworth projection of α-D-glucose is

(A) 

(B) 

(C) Both (A) and (B)

(D) None of these

28. Which of the following is having highest bondlength?

(A) NO^- (B) NO^+

(C) CN^- (D) CN^+

29. A 0.001 molal solution of $Pt(NH_3)_4Cl_4$ in aqueous solution decreases the freezing point by 0.0054^oC. If $K_f = 1.86$ K kg mol^{-1}. What is the formula of the complex:

(A) $[Pt(NH_3)_4Cl_4]$ (B) $[Pt(NH_3)_5Cl]Cl_3$

(C) $[Pt(NH_3)_4Cl_2]Cl_2$ (D) $[Pt(NH_3)_3Cl_3]Cl.NH_3$

30. Holm's signal can be given by using

(A) $CaC_2 + CaCN_2$ (B) $CaC_2 + Ca_3P_2$

(C) $CaC_2 + CaCO_3$ (D) $Ca_3P_2 + CaCN_2$

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The points A(3, 4) and B(4, 3) lies on same or opposite side of the line $ax + by + c = 0$, $(a, b, c \in R^+)$ in which origin lies and P_1 and P_2 are length of perpendicular from A and B to the line such that $2P_1 + 3P_2 = 10$ then the line $ax + by + c = 0$ touches the circle

(A) $(x - 18)^2 + (y - 17)^2 = 4$ (B) $\left(x + \frac{18}{5}\right)^2 + \left(y - \frac{17}{5}\right)^2 = 4$

(C) $\left(x - \frac{18}{5}\right)^2 + \left(y - \frac{17}{5}\right)^2 = 4$ (D) $(x - 3)^2 + (y - 6)^2 = 4$

2. If a hyperbola whose foci are (–2, 4) and (4, 6) touches y–axis then equation of hyperbola is

(A) $(3x + y - 8)^2 - \frac{(x - 3y + 14)^2}{4} = 20$ (B) $\frac{(x + 3y - 7)^2}{2} - \frac{(x - 3y + 8)^2}{8} = 1$

(C) $\frac{(3x + y - 8)^2}{2} - \frac{(x - 3y + 14)^2}{8} = 1$ (D) $\frac{(x - 3y + 14)^2}{2} - \frac{(3x + y - 8)^2}{8} = 20$

3. $f(x) = \frac{x^4 - 14x^2 + 25}{x - \frac{5}{x} + 2}$, zeroes are of the form $a \pm \sqrt{b}$, $a, b \in z$ then $a + b$ is equal to

(A) 5 (B) 6
(C) 7 (D) 8

4. The number 916238457 is an example of nine digit number which contain each of the digits 1 to 9 exactly one, it also has the property that the digit 1 to 5 occur in their natural order, while the digit 1 to 6 do not. Number of such numbers are

(A) 2268 (B) 2520
(C) 2975 (D) 1560

Space for rough work

5. Let $z = 18 + 26i$, where $z = a + ib$ $(a, b \in R)$ is the cube root of z having least positive argument then the value of $ab(a + b)$ is
(A) 18 (B) 12
(C) 10 (D) $\sqrt{10}$

6. A parabola touches two given straight lines originating from a given point. The locus of mid–point of portion of any tangent. Which is intercepted between the given straight line is
(A) parabola (B) ellipse
(C) straight line (D) hyperbola

7. z_1, z_2 and z_3 are three points on a circle centred at origin. A point z is chosen on the circle such that the line joining z and z_1 is perpendicular to the line joining z_2 and z_3. Which of the following is true?
(A) $zz_1 + z_2z_3 = 0$ (B) $z^2 - z_1^2 + z_2z_3 = 0$
(C) $z^2 + z_1^2 + z_2z_3 = 0$ (D) $zz_1 - z_2z_3 = 0$

8. Let $C_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)}\,dx$ then $\lim_{n\to\infty} n^2 \cdot C_n$ equals
(A) 1 (B) 0
(C) –1 (D) $\frac{1}{2}$

9. We have 21 identical balls available with us which we need to be distributed amongst 3 boys A, B and C such that A always gets an even number of balls. The number of possible ways of doing this is
(A) 112 (B) 120
(C) 126 (D) 132

10. The equation $x a^{1/x} + \frac{1}{x} a^x = 2a$, $a \geq 1$; $a, x \in R$ has
(A) only one solution as x = 1 (B) only positive solution as x = 1
(C) infinite solutions (D) none of these

Space for rough work

11. A card from a pack of 52 cards is lost. From the remaining cards two cards are drawn at random and are found to be spades. What is the probability that the missing card was a spade

(A) $\frac{1}{5}$ (B) $\frac{11}{50}$

(C) $\frac{6}{25}$ (D) $\frac{1}{4}$

12. The latus rectum of a conic section is the width of perpendicular line segment through the focus. The positive difference between the lengths of the latus rectum of $3y = x^2 + 4x - 9$ and $x^2 + 4y^2 - 6x + 16y = 24$ is

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{3}{2}$ (D) $\frac{5}{2}$

13. Let A be a $(n \times n)$ matrix with $|A| = 4$, B is the adjoint of the matrix 2A such that $|B| = 1024$. What is the value of n?

(A) 3 (B) 4

(C) 5 (D) more than 5

14. Suppose that the domain of the function f(x) is set D and the range is the set R, where D and R are the subsets of real numbers. Consider the functions: f(2x), f(x + 2), 2f(x), $f\left(\frac{x}{2}\right)$, $\frac{f(x)}{2} - 2$. If m is the number of functions listed above that must have the same domain as f(x) and n is the number of functions that must have the same range as f(x) then the ordered pair (m, n) is

(A) (1, 5) (B) (2, 3)

(C) (3, 2) (D) (3, 3)

15. Let $\vec{a}_1$, $\vec{a}_2$ ..... $\vec{a}_n$ be the sides of a regular polygon inscribed in a circle of unit radius. If $\left[\vec{a}_1 \times \vec{a}_2 + \vec{a}_2 \times \vec{a}_3 ..... + \vec{a}_n \times \vec{a}_1\right] = \left[\vec{a}_1 \cdot \vec{a}_2 + \vec{a}_2 \cdot \vec{a}_3 + ... + \vec{a}_n \cdot \vec{a}_1\right]$ smallest possible value of n is

(A) 4 (B) 6

(C) 8 (D) 12

Space for rough work

16. Let us consider $f(x) = 1 - x + [x] - [1 - x]$ and $g(x) = 1 - x - \lim_{n\to\infty} (\cos^{2n+1} \pi x)$

(A) the graph of f(x) is a straight line passing through origin.
The graph of g(x) is straight line passing through origin
(B) the graph of both f(x) and g(x) is a straight line not passing through origin
(C) the graph of f(x) is a straight line passing through origin whereas for that of g(x) is not passing through origin
(D) will not be a straight line for either of the curve

17. Let $f(x) = [x] + \{x\}^2 : R \to R$, then area of figure bounded by $y = f^{-1}(x)$, $y = 0$ between the ordinates $x = \frac{1}{2}$ and $x = 5$ is (where [.] and {.} represent the greatest integers and fractional part functions respectively)

(A) $\frac{1}{3\sqrt{2}}\left[40\sqrt{2}+1\right]$ (B) $\frac{40}{3}$

(C) $\frac{40}{3\sqrt{2}}$ (D) $\frac{1}{3\sqrt{2}}\left(40\sqrt{2}-1\right)$

18. What is the minimum value of $\frac{\sec^4\alpha}{\tan^2\beta}+\frac{\sec^4\beta}{\tan^2\alpha}$ for $\alpha, B \neq \frac{k\pi}{2}$, $k \in z$?

(A) 2 (B) 4
(C) 8 (D) 16

19. Consider straight line $ax + by = c$ where $abc \in R^+$ and a, b, c are distinct. This line meets the coordinate axes at P and Q respectively. If area of ΔOPQ, 'O' being origin does not depend upon a, b and c then

(A) a, b, c are in GP (B) a, c b are in GP
(C) a, b, c are in AP (D) a, c, b are in AP

20. In a ΔABC, a, c, $\angle A$ are fixed. The third side may have two possible values say b_1 and b_2. It is given that $b_2 = 2b_1$. Then find the value of $\frac{c}{a}\sqrt{1+8\sin^2 A}$

(A) 1 (B) 2
(C) 3 (D) 4

21. If A is an invertible idempotent matrix and $B = 7A^7 + 6A^6 + 5A^5 + ... + A$ then $|B|$ is equal to

(A) 7 (B) 14
(C) 28 (D) 35

Space for rough work

22. Find the number of triangles formed by the lines represented by $x^3 - x^2 - x - 2 = 0$ and $x y^2 + 2 x y + 4 x - 2 y^2 - 4 y - 8 = 0$
(A) one (B) two
(C) three (D) none of these

23. Let $f(x) = \sin^3 x + \lambda \sin^2 x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $\lambda > 0$. The interval in which λ should lie in order that f(x) has exactly one minimum and one maximum is
(A) $0 < \lambda < \frac{3}{2}$ (B) $0 < \lambda < 1$
(C) $1 < \lambda < 2$ (D) $1 < \lambda < 3/2$

24. If $P(x) = ax^2 + bx + c$ leaves a remainder of 4 when divided by x, a remainder of 3 when divided by (x + 1) and a remainder of 1 when divided by (x – 1) then P(2) is
(A) 3 (B) 6
(C) –3 (D) –6

25. Let $L_1 : 2x + 3y + p - 3 = 0$ and $L_2 : 2x + 3y + p + 3 = 0$ be two lines and $p \in z$.
Let $C : x^2 + y^2 + 6x + 10y + 30 = 0$. If it is given that at least one of the lines L_1 L_2 is chord of C the probability that both are chords of C is
(A) $\frac{2}{7}$ (B) $\frac{3}{7}$
(C) $\frac{4}{11}$ (D) $\frac{5}{11}$

26. If $\lim\limits_{n\to\infty} \frac{n \cdot 3^n}{n(x-2)^2 + n \cdot 3^{n+1} - 3^n} = \frac{1}{3}$ then the range of x is ($n \in N$)
(A) [2, 5] (B) (1, 5)
(C) (–1, 5) (D) $(-\infty, \infty)$

Space for rough work

27. The coordinate plane is constrained by the hyperbola $x^2 - y^2 = 4$ so that only the region exterior of the hyperbola is accessible what are the possible values of the ordinates of the center of a circular disc of radius 4 where centres lies on the y–axis?

(A) $\left(-\infty,\ -\sqrt{2}\right] \cup \left[\sqrt{2},\ \infty\right)$ (B) $\left(-\infty,\ -2\sqrt{3}\right] \cup \left[2\sqrt{3},\ \infty\right)$

(C) $\left(-\infty,\ -\sqrt{6}\right) \cup \left[\sqrt{6},\ \infty\right)$ (D) $\left(-\infty,\ -2\sqrt{6}\right] \cup \left[2\sqrt{6},\ \infty\right)$

28. If $\left|x^2 - \frac{\pi}{4} + \sin x - 2\right| = |x^2 - 2| + |\sin x| + \frac{\pi}{4}$ then

(A) $x \in (0,\ \sqrt{2})$ (B) $x \in (-\sqrt{2},\ \sqrt{2})$

(C) $x \in R$ (D) $x \in (-\sqrt{2},\ 0)$

29. A plane P makes intercepts with the axes, the sum of whose square is a constant equal to K^2. The foot of the perpendicular from the origin P is $(\lambda, \lambda, \lambda)$ the value of λ in terms of K is

(A) $\pm\frac{K}{3\sqrt{3}}$ (B) $\pm\frac{K}{2\sqrt{3}}$

(C) $\pm\frac{K}{\sqrt{3}}$ (D) $\pm\frac{2K}{\sqrt{3}}$

30. Let $f(x) = \ln\left(\frac{x^2 + e}{x^2 + 1}\right)$ then the range of $g(x) = \sqrt{\sin f(x)} + \sqrt{\cos f(x)}$ is

(A) $\left(1,\ 2^{3/4}\right]$ (B) $\left[1,\ 2^{1/2}\right)$

(C) $\left[1,\ 2^2\right]$ (D) (1, 2)

Space for rough work